

Chapter 02

Core Controls and Layout Fundamentals

1 lesson • 25 hr

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Chapter 02 • Core Controls and Layout Fundamentals

Practical project of the chapter

15 pages

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Core Controls and Layout Fundamentals

Driving Question

Build a single interactive card that serves as a preview of a journal entry, utilizing layout containers, input controls, and event handlers.

Production Format

Argumentative essay — 800 to 1500 words.

Context and Provocation

Objective

Build a single interactive card that displays a journal entry preview using layout containers, display controls, input controls, and buttons. This project exercises Row, Column, Stack, Container, Card, Text, Icon, Image, TextField, Dropdown, ElevatedButton, and basic event handling.

Creative Brief

Your friend who wanted the diary now says: "I want to see what each entry will look like before I save it. Can you make a little preview card? I want to be able to pick a mood from a dropdown, write a short title, and see the card update instantly. Oh, and a delete button that does nothing for now, just to see it." So you're building a prototype of a single journal entry card with live preview. This is a standalone component that will later be reused in the full journal app.

Objective

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Materials and Resources

- Python 3.10 or higher installed
- Flet library installed ('pip install flet')
- A code editor (VS Code, PyCharm, or similar)
- Internet access to the Flet documentation for icons and controls
- Optional: a local image file to use as an avatar (or use a network image like `'https://picsum.photos/100/100'`)

Execution Steps

1. Set up the main page with a vertical centered layout.
 - Professional insider tip: Use `'page.horizontal_alignment = "center"'` and wrap your card in a Column with `'alignment="center"'` and `'expand=True'`.
 - Common mistake to avoid: Forgetting to call `'page.update()'` after changes; always use `'set_state()'` inside event handlers.
2. Create a Card containing a Container with a Column. Inside the Column, add a Row for the top section: a TextField for the title and a Dropdown for mood.
 - Professional insider tip: Use `'expand=True'` on the TextField so it fills available space. For the Dropdown, set `'options'` with `ft.dropdown.Option` for each mood (e.g., "Happy", "Sad", "Excited", "Calm").
 - Common mistake to avoid: Putting the Dropdown inside a Row without proper alignment; use `'alignment="spaceBetween"'` to separate the two elements.
3. Add a second Row for the body: an Image control (network or local) and a multiline TextField for entry content.
 - Professional insider tip: Use `'ft.Image(src="https://picsum.photos/100/100", width=80, height=80, fit="cover")'` for a placeholder. Set the TextField `'multiline=True'` and `'min_lines=3'`.
 - Common mistake to avoid: Not setting a fixed width/height for the image, which may cause layout shifts.
4. Add a third Row at the bottom with an ElevatedButton labeled "Delete" (disabled) and an IconButton (e.g., `ft.icons.EDIT`).
 - Professional insider tip: Use `'disabled=True'` on the delete button to indicate it's not functional yet. The IconButton can have an `'on_click'` that prints a message for now.
 - Common mistake to avoid: Placing the Row directly below the body without proper spacing; add `'spacing=10'` to the Column.
5. Implement live preview: when the user types in the title TextField, update a Text control inside the card (e.g., a title text). When the mood Dropdown changes, update an emoji or icon next to the title. Use `'on_change'` events.
 - Professional insider tip: Store the title Text control as a reference (e.g., `'self.title_display'`) and update it inside the event handler with `'self.title_display.value = e.control.value'` and then `'self.page.update()'`. For

mood, create a Text control that shows an emoji like "?".

- Common mistake to avoid: Forgetting to update the page after changing values; always call 'self.page.update()' or use 'set_state()'.

Self-Assessment Checklist

- The card contains a title input that updates a displayed title in real time.
- The mood dropdown offers at least 3 options and changes an emoji or text indicator.
- An image placeholder (network or local) is visible on the left side.
- A multiline content area is present and scrollable.
- A delete button (disabled) and an edit icon button are at the bottom.
- The overall layout uses proper nesting of Rows and Columns.
- The card has a clean, card-like appearance (shadow, border radius).

Bonus Challenge

Add a Switch control that toggles the card's background color between light and dark mode. When the switch is flipped, change the Container's 'bgcolor' property accordingly.

Submission Format

Accepted formats:

- image (.png/.jpg/.webp)

Minimum delivery:

- One screenshot (.png) showing the card with a title typed, a mood selected, and the content area filled with a short sentence.

Development Guide

1. Research Phase

Look for examples of small apartment designs on sites like ArchDaily or Dezeen. Read news articles on micro-apartments to see both praise and criticism. Note how different designs handle the space-saving vs. comfort tradeoff. Use the course chapters on apartment planning and 3D modeling as your main guides.

2. Organization Phase

Start by stating your main argument — your position on balancing efficiency and comfort. Then outline three to four key points that support your view, using the mandatory references. For each point, plan what evidence or examples you'll use. Also think about one counterargument and how you'll respond to it.

3. Writing Phase

Write an introduction that presents the problem and your thesis. In the body paragraphs, each should focus on one key point — explaining a course concept and applying it to your design example. Use simple language: imagine explaining to a friend. Include specific examples, like how you'd place a bed or choose lighting. End with a conclusion that summarizes your argument and suggests practical tips for designers.

4. Review Phase

Check that your essay clearly states your position. Verify that you've used all four mandatory references correctly. Read it aloud to see if the logic flows smoothly. Ask yourself: would a beginner understand this? Remove any jargon or complex terms without explanation.

Self-Assessment Rubric

1. Title Live Update

- Insufficient — Title does not update, or updates only after dropdown change, or no separate display control exists.
- Adequate — Title updates on each keystroke, but maybe slight lag or displayed title is plain. Still works.
- Excellent — Title TextField updates a separate Text control in real time, character by character, with no delay. The displayed title is styled (e.g., bold, larger font).

2. Mood Emoji Update

- Insufficient — Dropdown missing, or no emoji displayed, or fewer than 3 options.
- Adequate — Dropdown has 3+ options, emoji changes but maybe one option missing, or emoji too small.
- Excellent — Dropdown has at least 3 mood options; each changes a visible emoji (e.g., 🤔, 😊) next to the title. Emoji is appropriately sized and clear.

3. Image Presence

- Insufficient — No image, or broken image (shows placeholder icon).
- Adequate — Image present but may be stretched or large. Still loads.
- Excellent — An image (local or network) is displayed with fixed size (e.g., 80x80) and 'fit="cover"'. Loads without broken icon.

4. Multiline Content Area

- Insufficient — Single-line only, or cannot type beyond one line.
- Adequate — Multiline works but 'min_lines' not set; still scrollable.
- Excellent — A multiline TextField with 'min_lines=3' that scrolls when content overflows. User can type multiple lines.

5. Action Buttons

- Insufficient — Delete not disabled, or one button missing.
- Adequate — Delete disabled, edit button exists but no handler.
- Excellent — Delete button disabled and labeled "Delete". Edit IconButton present with an 'on_click' handler (e.g., prints). Both are aligned at bottom row.

6. Layout & Structure

- Insufficient — Layout breaks: elements overlap, missing Card, or no Row/Column structure. Page not centered.
- Adequate — Layout works but may have cramped spacing or alignment issues. Still uses Card and Column/Row.
- Excellent — Card uses proper nesting: Card > Container > Column > Rows with appropriate spacing (e.g., spacing=10). Alignment is correct (e.g., spaceBetween). Card has shadow and rounded corners. Page is centered.

7. Bonus: Dark Mode Toggle

- Insufficient — No Switch, or switch does nothing.
- Adequate — Switch present but toggling may require extra click.
- Excellent — Switch control toggles card background between light and dark. Works immediately with 'page.update()'.

Model Response & Assessment Reference

This guide presents possible paths, not single answers. An excellent essay may defend any of the theses below — or an original thesis not anticipated here — as long as it demonstrates argumentative rigor, correct use of sources, and autonomous critical reflection.

1. Expected Thesis Position(s)

The central question admits multiple legitimate positions. Below are argumentative paths representing robust and distinct directions.

Author note: *The recommended range is 2 to 4 theses. If the assignment has only one defensible answer, keep just Thesis A and adapt the wording. A third thesis often works best as a synthetic or dialectical position.*

1.1. Thesis A — {Short label}

{Full thesis statement, in one or two sentences. It should take a clear position on the central question.}

Main argumentative line: {One paragraph describing how this thesis sustains itself: the central claim, supporting reasoning, implicit framework, and what makes it defensible.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

1.2. Thesis B — {Short label}

{Full thesis statement — a meaningfully different position from Thesis A, not just a softer variant.}

Main argumentative line: {One paragraph describing the central claim, supporting reasoning, implicit framework, and what makes it defensible.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

1.3. Thesis C — {Synthetic/dialectical position, optional}

{Full thesis statement — a meaningfully different position from Thesis A, not just a softer variant.}

Main argumentative line: {Show explicitly how this position produces a new insight that neither A nor B alone can reach.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

2. Key Concepts and Their Correct Use

Author note: Include 3 to 5 key concepts per project. For each concept: precise definition + example of correct application + common pitfall. The pitfall section is critical — it tells the evaluator what to watch for.

2.1. A — {Concept Name}

How to define/explain: {One paragraph defining the concept precisely. Include its theoretical origin or author when relevant. Define its scope and limits.}

Correct use (example): {A verbatim example paragraph showing the concept doing actual argumentative work — not just being named, but actively grounding a claim.}

Common pitfall: {The most frequent way students misuse this concept: over-application, decorative citation, missing nuance, ignoring critiques, or conflating related concepts. Be concrete.}

2.2. B — {Concept Name}

How to define/explain: {One paragraph defining the concept precisely. Include its theoretical origin or author when relevant. Define its scope and limits.}

Correct use (example): {A verbatim example paragraph showing the concept doing actual argumentative work and grounding a claim.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

2.3. C — {Concept Name}

How to define/explain: {One paragraph defining the concept precisely. Include its theoretical origin or author when relevant. Define its scope and limits.}

Correct use (example): {A verbatim example paragraph showing the concept doing actual argumentative work and grounding a claim.}

Concepts mobilized: {Course concept 1} ({brief connection}); {Course concept 2} ({brief connection}); {Course concept 3} ({brief connection}).

3. Model Argumentative Structure

The outline below illustrates the structure of an excellent-level essay, regardless of which thesis the student chooses.

Author note: Word ranges are suggestions calibrated for a {total length}-word essay. Adjust proportionally if the assignment is shorter or longer.

3.1. Introduction ({X}–{Y} words)

Contemporary hook: {Describe what kind of opening works for this project — a concrete example, striking data point, scene from culture, or current event. Adapt to the project's domain.}

Bridge to subject: {How the hook should pivot into the project's central question.}

Explicit thesis: The author's position in one or two clear, specific sentences, positioned directly against the central question.

3.2. Development ({X}–{Y} words)

Suggested: 2–3 paragraphs, each with a central argument.

Paragraph 1: {Function and how it advances the thesis.}

Paragraph 2: {Function and how it advances the thesis.}

Paragraph 3: {Function and how it advances the thesis.}

3.3. Counterargument ({X}–{Y} words)

An excellent essay does not merely mention an objection: it presents the strongest counterargument with intellectual honesty and responds consistently.

Author note: *List the best counterargument for each thesis option, with a suggested response. The student should not feel that confronting the objection weakens their position — done well, it strengthens it.*

If thesis is A: {Strongest objection and possible response.}

If thesis is B: {Strongest objection and possible response.}

If thesis is C: {Strongest objection and possible response.}

3.4. Conclusion ({X}–{Y} words)

Restatement of thesis: Reaffirm the position in light of the full argumentation, without repeating the introduction verbatim.

Implications: {What this reflection reveals about something larger — the field, how we think, or a contemporary issue.}

Opening (no new information): A provocative question or forward-looking gesture that broadens the horizon without introducing arguments not developed in the essay.

3.5. Evidence and Data Types to Mobilize

- Specific passages from primary sources.
- Theoretical concepts with citation to the original work.
- Documented contemporary examples: news, official reports, data, or cultural productions.
- References to academic debates covered in the course.

4. Rubric — Excellent Level

Detailed description of what the evaluator should observe in an essay rated “Excellent” on each criterion.

Author note: *The criteria below cover the standard dimensions of argumentative essays. Remove or adapt criteria that do not apply to the specific project.*

4.1. Thesis clarity

What is observed at the EXCELLENT level: The thesis appears in the introduction unequivocally. It is specific and directly engages the project's central tension. The entire essay is organized around it.

4.2. Use of course concepts

What is observed at the EXCELLENT level: Required references are defined precisely before application. Concepts are integrated organically into the argument, and their limits are recognized.

4.3. Quality of argumentation

What is observed at the EXCELLENT level: Each paragraph opens with a clear claim, sustains it with evidence, and connects back to the thesis. Logical progression is clear and the student analyzes rather than merely describes.

4.4. Counterarguments

What is observed at the EXCELLENT level: The student presents the strongest objection to their position and responds with a substantial argument. The thesis emerges stronger from confronting the critique.

4.5. Originality of analysis

What is observed at the EXCELLENT level: The essay goes beyond reproducing course materials. The student offers original connections, asks new questions, or proposes a novel interpretation from familiar evidence.

4.6. Connection to the contemporary

What is observed at the EXCELLENT level: The dialogue between subject and present is specific and grounded. Concrete contemporary examples illuminate both sides.

4.7. Structure and academic norms

What is observed at the EXCELLENT level: Cohesive structure with fluid transitions. Citations follow the established academic pattern, and references are complete and correctly formatted.

5. Common Argumentative Errors

The errors below are the most frequent in this type of project. The evaluator should watch for them.

Author note: Errors 2, 3, and 4 apply to almost any argumentative project. Errors 1 and 5 are project-specific slots where particular risks may be described.

5.1. {Project-specific error}

How it appears in the text: {Verbatim-style example showing the error in action.}

How to correct: {Concrete, actionable advice on what to do differently.}

5.2. Strawman fallacy

How it appears in the text: The student presents the counterargument in its weakest, easiest-to-refute form instead of confronting the strongest version.

How to correct: Reformulate the counterargument as an intelligent interlocutor would present it, then respond to that reason.

5.3. Decorative use of concepts

How it appears in the text: The student names a concept but never applies it — it appears as a loose citation, disconnected from the argument.

How to correct: After citing a concept, explain exactly how it grounds the specific argument of that paragraph.

5.4. False equivalence

How it appears in the text: The student treats as equivalent two phenomena that operate in radically different contexts or functions.

How to correct: Make differences explicit before drawing parallels. Use precise comparative language. Comparison is not equation.

5.5. {Project-specific error}

How it appears in the text: {Example}

How to correct: {Correction}

Final reminder: *The quality of reasoning is always more important than the position defended. An essay that defends an unpopular thesis with rigor outperforms one that defends a consensus thesis superficially.*

